Simple Machines – Test Revision

What have you learnt?

Remembering

1 **Identify** an example of each type of simple machine.

Understanding

2 Draw a diagram to illustrate the way a slope works to lift things up high.

3 **Outline** 10 places where you would see a screw at work (such as the lid of a jar).

Applying

4 Use a diagram to **identify** the load, effort and pivot on a wheelbarrow.

5 **Describe** how can you pull downwards, yet lift something up.

6 **Outline** the use of a machine that doubles the speed of rotation of a toothed wheel.

Analysing

7 **Outline** 10 things you could not do if there were no wheels.

8 **Describe** which gear you would choose to pedal a bicycle up a hill. Why would you choose this gear?

9 **Describe** which gear you would choose to pedal a bike on a long flat surface. Why would you choose this gear?

10 **Evaluate** how a person with a physical disability uses one simple machine to assist them.

Machines and work

1 Examine Figure 7.6 to answer the following.

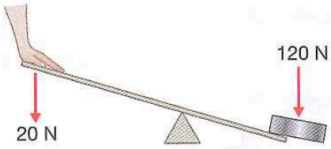


Fig 7.6

a What type of simple machine is this? _____

b Is it acting as a force multiplier or a distance multiplier? _____

c Calculate the mechanical advantage. _____

2 Examine Figure 7.7 to answer the following.

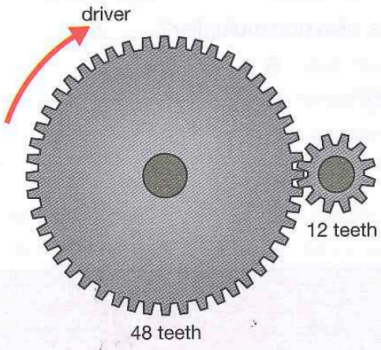


Fig 7.7

a What type of simple machine is this? _____

b Is it acting as a force multiplier or a distance multiplier? _____

c Calculate the mechanical advantage. _____

_____

3 Examine Figure 7.8 to answer the following.

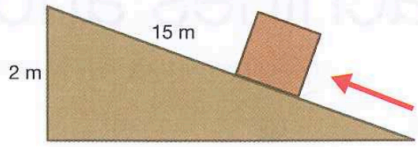


Fig 7.8

a What type of simple machine is this? _____

b Is it acting as a force multiplier or a distance multiplier? _____

c Calculate the mechanical advantage. _____

_____

4 Examine Figure 7.9 to answer the following.

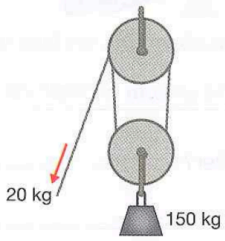


Fig 7.9

a What type of simple machine is this? _____

b Is it acting as a force multiplier or a distance multiplier? _____

c Calculate the mechanical advantage. _____

_____

5 Complete the table below.

Machine classification	Is the load force increased or decreased by the machine?	Is the distance moved by load increased or decreased by the machine?	Mechanical advantage greater than 1 or less than 1?
Force multiplier			
Distance multiplier			
Direction changer			